CLINICAL STUDY PROTOCOL

Amendment 1 — Study KLV-7702

Kelvaparib (KLV-3308)

Marnholt Oncology GmbH

Indication: Advanced solid tumours with a homologous recombination repair defect

Table A Amendment summary

Item	Detail
Protocol number	KLV-7702
Amendment number	1
Protocol version	2.0
Amendment date	1 September 2017
Supersedes	Version 1.0
Phase	Phase 1
Sponsor	Marnholt Oncology GmbH
Applies to	All participating centres
Substantial amendment	Yes
Document status	Complete replacement protocol

A1. REASON FOR THIS AMENDMENT

Clarification of eligibility and monitoring.

This document is a complete replacement for the previous version. The changes introduced by this amendment are listed in Section A2; the remainder of the document is the protocol as it now stands, so that sites work from one text rather than from a protocol plus a stack of change notices.

A2. CHANGES INTRODUCED BY THIS AMENDMENT

- Added an explicit requirement for full blood count before treatment, weekly for the first month and monthly thereafter, with dose interruption for grade 3 or higher cytopenia and prompt haematology referral for any cytopenia persisting beyond four weeks.
- Clarified the definition of the safety analysis set to include any subject receiving at least one dose.

A3. HISTORY OF AMENDMENTS TO THIS PROTOCOL

Version	Amendment	Date	Principal change
1.0	—	17 October 2016	Original approved version.
2.0	1	1 September 2017	Clarification of eligibility and monitoring

A4. EFFECT ON SUBJECTS ALREADY ENROLLED

Subjects already enrolled continue on study treatment without interruption. Those affected by a revised monitoring requirement are brought into line at their next scheduled visit. Re-consent is not required, because the changes do not alter the risks described in the subject information sheet.

A5. EFFECT ON THE STATISTICAL ANALYSIS

The primary analysis is unchanged. No change to the analysis populations or to the primary endpoint is introduced by this amendment.

A6. REGULATORY STATUS

This amendment is implemented only after approval by the responsible regulatory authorities and ethics committees in each territory, except where a change is necessary to eliminate an immediate hazard to a subject. The date of implementation at each site is recorded in the trial master file.

PROTOCOL AS AMENDED — VERSION 2.0

The complete protocol follows. It incorporates every change listed in Section A2 and every change made by the earlier amendments listed in Section A3.

SYNOPSIS

Title	A Phase 1 Open-label, Dose-escalation Study of Kelvaparib in Adults with Advanced Solid Tumours
Protocol number	KLV-7702
Phase	Phase 1
Sponsor	Marnholt Oncology GmbH
Primary objective	Maximum tolerated dose and recommended phase 2 dose
Study design	An open-label, multicentre, non-randomised, sequential dose-escalation study using a 3+3 design, followed by expansion at the dose selected as the recommended phase 2 dose. Approximately 58 subjects across 4 dose cohorts.
Population	Advanced solid tumours with a homologous recombination repair defect
Planned sample size	Approximately 58 subjects.
Investigational product	Kelvaparib, film-coated tablets, administered orally twice daily on a continuous schedule, with weekly blood counts for the first month and monthly thereafter.
Duration of treatment	Until disease progression, unacceptable toxicity or withdrawal of consent.
Primary endpoint	Incidence of dose-limiting toxicity during the first 21 days at the assigned cohort dose, and the maximum tolerated dose derived from it.
Key secondary endpoints	Duration of response, progression-free survival, overall survival and the incidence and severity of treatment-emergent adverse events.
Statistical methods	No formal hypothesis testing is planned. Sample size follows the operating characteristics of the 3+3 design and the number of subjects required to characterise safety at the selected dose. Proportions are presented with exact Clopper-Pearson 95 per cent confidence intervals and time-to-event endpoints estimated by the Kaplan-Meier method.

1 INTRODUCTION

1.1 Background to the disease

Advanced solid tumours with a homologous recombination repair defect are the indication under study in this protocol. The population enrolled here has disease that has progressed after, or is unsuitable for, the therapy ordinarily offered first, and the outcomes reported for that population remain poor. Treatment at this stage is given with the intention of controlling disease, delaying deterioration and preserving function, and the size of that benefit with existing options is modest. There is therefore a recognised need for agents that act by a mechanism not already exploited in earlier lines.

The natural history of the disease and the outcomes achieved with current management were reviewed when this protocol was written. That review informed the choice of a single-arm design, the endpoints, the assessment schedule and the assumptions underlying the sample size in Section 8.1.

The population studied in this protocol has advanced solid tumours with a homologous recombination repair defect. Disease at this point in the pathway behaves differently from disease at presentation: it has already been selected by exposure to prior therapy, and the mechanisms that allowed it to persist through that therapy are the mechanisms that determine what happens next. That selection is why response rates fall and why the duration of any response shortens with each successive line.

The clinical course is dominated by the accumulation of disease-related morbidity alongside the cumulative toxicity of previous treatment. Subjects entering a study at this stage typically carry a burden of unresolved effects from earlier lines, which constrains what can safely be given and which is the reason the eligibility criteria in Section 4 require recovery from prior toxicity before a subject may start study treatment.

Current management at this stage is directed at controlling disease, delaying deterioration and preserving function rather than at cure. The options available differ between territories and between institutions, and none of them is established as a standard against which a new agent must be compared. That variability is one reason the study has no concurrent comparator, as discussed in Section 3.2.

Outcomes reported for this population remain poor. Estimates vary with how the population was selected and with how long it was followed, so they are not directly comparable between series,

but the direction is consistent: response to further therapy is uncommon, responses that do occur are usually short, and the interval to further deterioration is measured in months. The assumptions used in the sample size calculation in Section 8.1 were chosen against that background.

The unmet need is therefore not simply for another agent but for an agent that acts by a mechanism the disease has not already been selected against. Kelvaparib is being developed on that basis, and the rationale for studying it in this population specifically is given in Section 1.2.

1.2 Background to the investigational product

Kelvaparib (KLV-3308) is an oral PARP1-selective inhibitor. Its mechanism is distinct from the agents used earlier in the treatment pathway, which provides the rationale for studying it in a population that has already been exposed to those agents.

The product is presented as film-coated tablets and is administered orally twice daily on a continuous schedule . Details of preparation, storage and handling are given in Section 5.2 and in the pharmacy manual.

Exposure after administration has been characterised in the studies described in the investigator brochure. The dose selected for this study, and the dose levels available for reduction in an individual subject, follow from that characterisation together with the relationship between exposure and toxicity; the reasoning is set out in Section 3.3.

The principal identified risk associated with this mechanism is myelodysplastic syndrome or acute myeloid leukaemia. It is predictable in its timing, it is detectable by the monitoring in Table 5 before it becomes clinically significant in most subjects, and it is manageable by the dose modification in Table 4 and the measures in Appendix 5. Those three facts together are what make it acceptable to study the product in this population.

Advanced solid tumours that have progressed after standard therapy carry a poor prognosis. Tumours with a homologous recombination repair defect are those in which PARP1 inhibition is expected to act.

Kelvaparib (KLV-3308) is an oral PARP1-selective inhibitor. By selectively inhibiting and trapping poly(ADP-ribose) polymerase 1 on damaged DNA it prevents single-strand break repair, so that replication forks collapse into double-strand breaks that tumours deficient in homologous recombination cannot repair. Because the effect is rapid and extensive, myelosuppression is a

recognised risk and is mitigated by the blood count monitoring and the dose modification rules in Section 5.6.

1.3 Non-clinical experience

Non-clinical pharmacology established activity at exposures achievable in the clinic and characterised the relationship between exposure and effect. Safety pharmacology found no effect on central nervous, respiratory or cardiovascular function at exposures within the intended clinical range. Repeat-dose toxicology in two species identified the target organs described in the investigator brochure; the findings were monitorable, and reversible on withdrawal at the exposures tested. Genotoxicity assays were negative. The full non-clinical package, including the derivation of the starting dose, is in the investigator brochure.

Primary pharmacology established activity in the relevant models at exposures that are achievable in the clinic, and characterised the relationship between exposure and effect. That relationship was used to define the exposure the clinical dose is intended to achieve.

Secondary pharmacology screening against a panel of receptors, ion channels, transporters and enzymes did not identify off-target activity at concentrations within the intended clinical range that would be expected to produce a clinical effect.

Safety pharmacology found no effect on central nervous system function, on respiratory function or on cardiovascular function, including cardiac repolarisation assessed in vitro and in vivo, at exposures within the intended clinical range. The electrocardiographic monitoring required by this protocol is nonetheless retained, because the non-clinical assessment cannot exclude an effect that emerges only in the presence of disease or of concomitant medication.

Repeat-dose toxicology in two species identified the target organs described in the investigator brochure. The findings were dose-related, were detectable by the laboratory monitoring applied in this protocol before they became severe, and were reversible on withdrawal at the exposures tested. The exposure margins between the no-observed-adverse-effect level and the intended clinical exposure are stated in the investigator brochure.

Genotoxicity assays were negative. Studies of reproductive and developmental toxicity, and of carcinogenic potential, are at the stage described in the investigator brochure; until they are complete the contraception requirements in Appendix 4 apply in full and are not open to local variation that would weaken them.

Absorption, distribution, metabolism and excretion were characterised in the non-clinical species and informed the interaction restrictions in Section 5.7. Where a clearance pathway is susceptible to inhibition or induction, the corresponding medication is prohibited or to be avoided, and the monitoring applied where it cannot be avoided is specified.

1.4 Clinical experience

Clinical experience with Kelvaparib at the time this protocol was written is summarised in the current edition of the investigator brochure and comprises the studies listed in the development plan. The safety profile observed to date is dominated by the events for which specific monitoring and management are required by this protocol; those events and their management are set out in Section 5.6 and Appendix 5.

Clinical experience with Kelvaparib at the time this protocol was written is summarised in the current edition of the investigator brochure. That edition is the reference safety information against which the expectedness of a serious adverse reaction is assessed, and investigators are notified whenever it is revised.

The events reported most often to date are those for which this protocol requires specific monitoring and management: nausea, anaemia, fatigue. Their incidence, their usual time of onset and their outcome are described in the investigator brochure and determine the assessment schedule in Table 5.

No finding in the clinical experience to date has required a restriction of the population beyond the eligibility criteria in Section 4, or a change to the dose modification rules in Table 4, other than those already incorporated in the current version of this protocol. The amendment history in Appendix 6 records where such a change was made and why.

1.5 Benefit and risk assessment

The anticipated benefit for a subject entering this study is the possibility of disease control in a setting where the available options give little. The principal identified risks are the adverse events of special interest listed in Section 7.3, together with the class effects expected of an oral PARP1-selective inhibitor. These risks are addressed by the eligibility criteria, by the monitoring schedule in Table 5, by mandatory dose modification in Table 4 and by the stopping rules in Section 3.5. In the judgement of the sponsor the balance of anticipated benefit against identified and potential risk supports conduct of the study as designed.

The anticipated benefit for an individual subject is the possibility of disease control in a setting where the available options offer little. That possibility is not assured, and the consent documentation in Appendix 16 states plainly that it is not.

The identified risks are myelodysplastic syndrome or acute myeloid leukaemia, pneumonitis, severe anaemia together with the class effects expected of an oral PARP1-selective inhibitor. The potential risks are those that would be expected from the mechanism but have not yet been observed, and those arising from the incompleteness of the non-clinical package described in Section 1.3.

Each identified risk is addressed by a specific measure: exclusion of subjects at disproportionate risk under Section 4.2, monitoring at the frequency in Table 5, mandatory withholding and reduction under Table 4, management guidance in Appendix 5, and reporting to the sponsor within 24 hours under Section 7.3 so that a change in the pattern is detected across the study rather than at one site.

In the judgement of the sponsor the balance of anticipated benefit against identified and potential risk supports conduct of the study as designed, in the population defined in Section 4, with the monitoring and the stopping rules described. That judgement is kept under review as data accumulate, and the arrangements for that review are in Section 3.5.

2 OBJECTIVES AND ENDPOINTS

Table 1 Objectives and endpoints

Type	Objective	Endpoint
Primary	To evaluate the safety and tolerability of Kelvaparib in subjects with advanced solid tumours with a homologous recombination repair defect.	Maximum tolerated dose and recommended phase 2 dose
Safety	To characterise the safety and tolerability of Kelvaparib.	Incidence, severity, seriousness and relationship to treatment of adverse events; clinically significant laboratory, vital sign and electrocardiogram changes.
Pharmacokinetic	To describe the pharmacokinetics of Kelvaparib in this population.	Plasma concentrations and the derived parameters listed in Section 7.4.
Exploratory	To explore relationships between candidate biomarkers and outcome.	Candidate biomarkers measured in blood and, where available, tissue.

2.1 Definition and derivation of each endpoint

Primary endpoint: Maximum tolerated dose and recommended phase 2 dose

Definition. The maximum tolerated dose is the highest dose level at which fewer than 33 per cent of DLT-evaluable subjects had a dose-limiting toxicity (Section 7.3.1); the recommended phase 2 dose is selected at or below it. It is assessed at the visits in Table 5. Where the endpoint depends on a determination that can be made by more than one party, the party whose determination is used for the analysis is stated in the statistical analysis plan.

Derivation. The endpoint is derived from the recorded assessments by the rules in the statistical analysis plan, which fix the handling of a missing assessment, of an assessment performed outside its window, and of an event occurring after the start of subsequent therapy. The derivation is programmed once and applied to every analysis of the endpoint.

Rationale. This endpoint was chosen because it can be measured reliably in this population at the interval the protocol allows, and because a change in it is interpretable as a change in the subject's condition rather than as an artefact of how often the subject was assessed.

Secondary endpoints. Progression-free survival is the time from the first dose of study treatment to the first documented disease progression by investigator assessment or death from any cause, whichever occurs first. Overall survival is the time from the first dose of study treatment to death from any cause. Duration of response is measured, for a subject with a confirmed complete or partial response, from the first dose of study treatment to the end of response recorded for the subject; a responder whose response has not ended is censored at the last assessment.

3 STUDY DESIGN

3.1 Overall design

An open-label, multicentre, non-randomised, sequential dose-escalation study using a 3+3 design, followed by expansion at the dose selected as the recommended phase 2 dose. Approximately 58 subjects across 4 dose cohorts.

The study is conducted at multiple centres. A subject's participation comprises a screening period of up to 28 days, a treatment period continuing until a discontinuation criterion in Section 6.1 is met, a safety follow-up visit 30 days after the last dose, and survival follow-up every 12 weeks thereafter.

The population studied here is advanced solid tumours with a homologous recombination repair defect. The design follows from that: the comparator, the endpoints and the assessment schedule are those appropriate to subjects at this point in the disease.

3.2 Rationale for the design

The design follows from the question the study is intended to answer. A single-arm design is used because the primary endpoint, dose-limiting toxicity in the first cycle, is observed within each subject and needs no concurrent control, and because randomisation is not practicable in a population of this size at this point in the pathway. The assessment schedule in Table 5 is set at the interval needed to detect the primary endpoint events without imposing avoidable burden on the subject.

The number of subjects is driven by the precision required of the primary estimate rather than by feasibility alone, and the derivation is in Section 8.1. Enrolment is spread across multiple centres and territories so that the population is not peculiar to one health system.

The duration of treatment is not fixed. A subject continues while deriving benefit and while tolerating treatment, which is the arrangement that best reflects how the product would be used, and which avoids stopping treatment in a subject who is benefiting merely because a protocol-defined period has elapsed.

Assessment is scheduled by calendar rather than by cycle for the disease assessments, so that a subject whose dosing is delayed is not also assessed late. This avoids the bias that arises when the subjects who are doing worst, and therefore have most interruptions, are assessed least often.

3.3 Rationale for the dose

The dose studied here is supported by the exposure achieved in earlier studies, by the relationship between exposure and toxicity observed in those studies, and by the non-clinical exposure margins. Dose levels available for individual reduction are in Table 3.

The dose was selected to achieve the exposure associated with activity in the non-clinical models while remaining below the exposure associated with the principal toxicity in the clinical experience to date.

No escalation above the starting dose is permitted for an individual subject. Reduction is permitted to the levels in Table 3, and re-escalation after a reduction is not permitted, because a subject who required a reduction has demonstrated sensitivity that is unlikely to have resolved.

The dosing schedule and the monitoring that accompanies it are in Section 5.6.

3.4 End of study definition

The end of the study is the date of the last visit, last scheduled procedure or last contact for survival of the last subject, whichever is later. The sponsor notifies the responsible authorities and ethics committees within the period required in each territory.

3.5 Stopping rules and study oversight

Enrolment is suspended pending review if the frequency or severity of the events of special interest exceeds the threshold defined in the safety monitoring plan, or if any event occurs that in the judgement of the sponsor alters the balance of benefit and risk. Oversight arrangements are described in the safety monitoring plan.

4 STUDY POPULATION

Approximately 58 subjects are planned. A subject is eligible only if every inclusion criterion in Section 4.1 is met and no exclusion criterion in Section 4.2 applies. Waivers of eligibility criteria are not permitted.

The criteria below define a population in which the question can be answered and in which the risk is acceptable. They are not a statement about who might benefit from the product if it were approved.

Every criterion is verified against source before a subject is enrolled, and the verification is documented. A subject who does not meet every inclusion criterion and who meets any exclusion criterion is a screen failure under Section 4.4, whatever the clinical judgement at the site.

4.1 Inclusion criteria

- Written informed consent obtained from the subject before any study-specific procedure is performed.
- Age 18 years or older at the time of consent.
- Histologically or cytologically confirmed advanced solid tumours with a homologous recombination repair defect.
- No remaining therapy of established benefit, or intolerance of or refusal of such therapy.
- Disease that is measurable or evaluable according to RECIST version 1.1.

- Eastern Cooperative Oncology Group performance status of 0 or 1, or 2 where the investigator judges the subject fit for study treatment.
- Life expectancy of at least 12 weeks in the opinion of the investigator.
- Adequate bone marrow function: absolute neutrophil count at or above 1.0 x 10^9/L, platelets at or above 75 x 10^9/L and haemoglobin at or above 80 g/L, without transfusion or growth factor support in the 7 days before screening.
- Adequate hepatic function: total bilirubin at or below 1.5 times the upper limit of normal, and alanine and aspartate aminotransferase at or below 3 times the upper limit of normal, or 5 times in the presence of hepatic involvement by the underlying disease.
- Adequate renal function: creatinine clearance at or above 40 mL/min estimated by the Cockcroft-Gault formula.
- Recovery from the acute toxic effects of prior therapy to grade 1 or baseline, other than alopecia and stable peripheral neuropathy.
- Women of childbearing potential must have a negative serum pregnancy test at screening and agree to use a highly effective method of contraception for the period defined in Appendix 4.
- Men must agree to use a barrier method and to refrain from donating sperm for the period defined in Appendix 4.
- Willing and able to comply with the schedule of assessments, the study restrictions and the requirements of long-term follow-up.

4.2 Exclusion criteria

- Any medical or psychiatric condition that in the opinion of the investigator would compromise the subject's safety or the interpretation of the study data.
- Treatment with any investigational agent within 28 days or five half-lives, whichever is shorter, before the first dose of study treatment.
- Major surgery within 28 days before the first dose, or a surgical procedure planned during the treatment period.
- Radiotherapy within 14 days before the first dose, other than a single fraction given for symptom control at a site that is not the target of assessment.
- Known hypersensitivity to Kelvaparib or to any excipient of the formulation.

- Active central nervous system involvement by the underlying disease, unless treated, stable for at least 28 days and not requiring corticosteroids above a physiological replacement dose.
- Uncontrolled intercurrent infection requiring systemic therapy at the time of the first dose.
- Known active hepatitis B or hepatitis C infection, or known human immunodeficiency virus infection with a detectable viral load.
- Clinically significant cardiovascular disease, including myocardial infarction or unstable angina within 6 months, uncontrolled arrhythmia, or heart failure of New York Heart Association class III or IV.
- A mean corrected QT interval above 470 ms on triplicate screening electrocardiogram, or a personal or family history of long QT syndrome.
- A history of myelodysplastic syndrome or acute myeloid leukaemia that in the opinion of the investigator places the subject at unacceptable risk of recurrence on study treatment.
- Any other malignancy that has required systemic treatment within 2 years, other than adequately treated non-melanoma skin cancer or carcinoma in situ.
- Pregnancy or breastfeeding, or intention to become pregnant during the study.
- Receipt of a live attenuated vaccine within 28 days before the first dose.
- Inability to swallow oral medication or a condition that would materially impair absorption.

4.2.1 Rationale for the eligibility criteria

Each criterion is given below with the reason it is there, so that a committee reviewing the protocol and an investigator applying it can see what each one is protecting against.

Inclusion criteria

1. Written informed consent obtained from the subject before any study-specific procedure is performed. Rationale. Consent precedes every study-specific act, so this criterion is verified first and its date is the anchor against which the timing of every screening assessment is checked.

2. Age 18 years or older at the time of consent. Rationale. The product has not been studied in people below the age of majority and the non-clinical package does not support it, so enrolment is restricted to adults.

3. Histologically or cytologically confirmed advanced solid tumours with a homologous recombination repair defect. Rationale. Confirmation on tissue or cytology, rather than on

imaging alone, is required because the primary endpoint is interpretable only in a population whose disease is what it is believed to be.

4. No remaining therapy of established benefit, or intolerance of or refusal of such therapy. Rationale. In an early-phase study the balance of benefit and risk is acceptable only where the alternative is no therapy of established benefit, or where the subject has declined or cannot tolerate it.

5. Disease that is measurable or evaluable according to RECIST version 1.1. Rationale. Without a lesion or parameter that can be followed, a subject cannot contribute to the primary endpoint, and enrolling such a subject would expose them to risk with no possibility of the study learning anything from them.

6. Eastern Cooperative Oncology Group performance status of 0 or 1, or 2 where the investigator judges the subject fit for study treatment. Rationale. Performance status is the best available single predictor of tolerance. The threshold admits subjects who are likely to tolerate treatment while excluding those for whom the balance of benefit and risk is unfavorable.

7. Life expectancy of at least 12 weeks in the opinion of the investigator. Rationale. A subject who is unlikely to survive long enough for the first scheduled assessment cannot contribute to the endpoint and would be exposed to the risks of treatment without the possibility of benefit.

8. Adequate bone marrow function: absolute neutrophil count at or above 1.0 x 10^9/L, platelets at or above 75 x 10^9/L and haemoglobin at or above 80 g/L, without transfusion or growth factor support in the 7 days before screening. Rationale. The thresholds are set so that a subject has the reserve to tolerate the expected haematological effect. They are deliberately not so strict as to exclude subjects whose counts reflect disease rather than frailty.

9. Adequate hepatic function: total bilirubin at or below 1.5 times the upper limit of normal, and alanine and aspartate aminotransferase at or below 3 times the upper limit of normal, or 5 times in the presence of hepatic involvement by the underlying disease. Rationale. Hepatic impairment alters exposure and reduces the margin before hepatotoxicity becomes clinically significant. The allowance for disease involvement of the liver reflects that a raised transaminase in that setting has a different meaning.

10. Adequate renal function: creatinine clearance at or above 40 mL/min estimated by the Cockcroft-Gault formula. Rationale. The threshold reflects the contribution of renal elimination to

clearance and the exposure that would result at lower function. Subjects below it would receive an exposure the safety data do not cover.

11. Recovery from the acute toxic effects of prior therapy to grade 1 or baseline, other than alopecia and stable peripheral neuropathy. Rationale. Entering a study before recovery from prior therapy makes it impossible to attribute a subsequent event, and compounds toxicity in a way that neither the eligibility criteria nor the dose modification rules were designed to handle.

12. Women of childbearing potential must have a negative serum pregnancy test at screening and agree to use a highly effective method of contraception for the period defined in Appendix 4. Rationale. The reproductive toxicity package is incomplete, so the requirement is absolute and is verified at screening and repeated during treatment as scheduled in Table 5.

13. Men must agree to use a barrier method and to refrain from donating sperm for the period defined in Appendix 4. Rationale. The requirement for men reflects the interval over which the product and its metabolites may be present in seminal fluid, with a margin.

14. Willing and able to comply with the schedule of assessments, the study restrictions and the requirements of long-term follow-up. Rationale. A study of this intensity depends on attendance. This criterion is about the practicability of participation for the subject, not about their motivation, and it is assessed with them rather than about them.

Exclusion criteria

1. Any medical or psychiatric condition that in the opinion of the investigator would compromise the subject's safety or the interpretation of the study data. Rationale. This is the criterion that allows the investigator to exclude a subject whose particular circumstances make participation unsafe or uninterpretable but which no specific criterion anticipates. Its use is recorded with the reason.

2. Treatment with any investigational agent within 28 days or five half-lives, whichever is shorter, before the first dose of study treatment. Rationale. A washout of five half-lives or 28 days removes the possibility that an event is attributed to this product when it belongs to the previous one.

3. Major surgery within 28 days before the first dose, or a surgical procedure planned during the treatment period. Rationale. Recent surgery introduces a period of altered physiology and a wound that may be affected by the product, and it makes attribution of events difficult during the interval when the study is most closely monitoring for them.

4. Radiotherapy within 14 days before the first dose, other than a single fraction given for symptom control at a site that is not the target of assessment. Rationale. Recent radiotherapy produces changes at the treated site that can be mistaken for a change in disease. The exception for a single palliative fraction outside the assessed sites preserves access to symptom control.

5. Known hypersensitivity to Kelvaparib or to any excipient of the formulation. Rationale. Known hypersensitivity to the product or an excipient is an absolute contraindication.

6. Active central nervous system involvement by the underlying disease, unless treated, stable for at least 28 days and not requiring corticosteroids above a physiological replacement dose. Rationale. Untreated or unstable involvement carries its own risk of deterioration that would confound both safety and efficacy assessment, and the requirement for corticosteroid would interact with the product.

7. Uncontrolled intercurrent infection requiring systemic therapy at the time of the first dose. Rationale. An infection requiring systemic therapy at the time of first dose would be worsened by the expected effect on host defence and would make attribution of a subsequent event impossible.

8. Known active hepatitis B or hepatitis C infection, or known human immunodeficiency virus infection with a detectable viral load. Rationale. Active viral hepatitis alters the interpretation of hepatic laboratory abnormality and carries a risk of reactivation.

9. Clinically significant cardiovascular disease, including myocardial infarction or unstable angina within 6 months, uncontrolled arrhythmia, or heart failure of New York Heart Association class III or IV. Rationale. The specified conditions carry a short-term risk of an event that would be attributed to the product, and the monitoring required by this protocol is not designed to manage them.

10. A mean corrected QT interval above 470 ms on triplicate screening electrocardiogram, or a personal or family history of long QT syndrome. Rationale. Although the non-clinical assessment found no effect on repolarisation, exclusion of subjects with an already prolonged interval preserves the margin in a population that will receive concomitant medication.

11. A history of myelodysplastic syndrome or acute myeloid leukaemia that in the opinion of the investigator places the subject at unacceptable risk of recurrence on study treatment. Rationale. A subject who has already demonstrated susceptibility to the principal identified risk is at

disproportionate risk of recurrence, which the dose modification rules would not adequately mitigate.

12. Any other malignancy that has required systemic treatment within 2 years, other than adequately treated non-melanoma skin cancer or carcinoma in situ. Rationale. A concurrent malignancy requiring treatment confounds both the efficacy endpoint and the attribution of adverse events. The exceptions are conditions that neither require systemic treatment nor affect survival.

13. Pregnancy or breastfeeding, or intention to become pregnant during the study. Rationale. Absolute, for the reasons given in Section 1.3 and Appendix 4.

14. Receipt of a live attenuated vaccine within 28 days before the first dose. Rationale. The expected effect on host defence makes a live vaccine unsafe within the specified interval.

15. Inability to swallow oral medication or a condition that would materially impair absorption. Rationale. Because the product is oral, a subject who cannot reliably swallow or absorb it would receive an unknown exposure.

Eligibility is defined by the population stated above, together with the organ function and performance status thresholds set out in the inclusion criteria.

4.3 Lifestyle restrictions

- Subjects must not donate blood or plasma during the study or for 90 days after the last dose; men must not donate sperm during the study or for 120 days after the last dose (Appendix 4).
- Grapefruit, grapefruit juice and Seville oranges are prohibited from screening until 7 days after the last dose.
- Alcohol is discouraged during the treatment period and prohibited in the 24 hours before a pharmacokinetic sampling visit.
- Contraception requirements are in Appendix 4.

4.4 Screen failures

A subject who consents but is not enrolled is a screen failure. Demography, the reason for failure and any serious adverse event occurring after consent are recorded. A subject may be re-screened once where the reason for failure was transient; the subject retains the original identifier and the fact of re-screening is recorded.

4.5 Replacement of subjects

A subject who discontinues is not replaced, except that in the dose-escalation part a subject who is not evaluable for dose-limiting toxicity is replaced so that each cohort contains the required number of evaluable subjects.

5 STUDY TREATMENT

5.1 Treatments administered

Table 2 Treatments administered

Arm	Product	Dose and route	Schedule
100 mg	Kelvaparib	100 mg orally twice daily on a continuous schedule	Continuous; 28-day cycles
200 mg	Kelvaparib	200 mg orally twice daily on a continuous schedule	Continuous; 28-day cycles
300 mg	Kelvaparib	300 mg orally twice daily on a continuous schedule	Continuous; 28-day cycles
400 mg	Kelvaparib	400 mg orally twice daily on a continuous schedule	Continuous; 28-day cycles

Table 3 Dose levels

Dose level	Dose	Route and schedule
Cohort 1	100 mg	orally twice daily on a continuous schedule
Cohort 2	200 mg	orally twice daily on a continuous schedule
Cohort 3	300 mg	orally twice daily on a continuous schedule
Cohort 4	400 mg	orally twice daily on a continuous schedule
Dose level -1	75 per cent of the assigned cohort dose	Unchanged
Dose level -2	50 per cent of the assigned cohort dose	Unchanged

No dose escalation above the starting dose is permitted for an individual subject.

5.2 Preparation, handling, storage and accountability

Study treatment is supplied by the sponsor, labelled in accordance with the requirements of each territory, and stored under the conditions on the label with continuous temperature monitoring. Preparation is performed by a pharmacist or other authorised person following the pharmacy manual. The investigator is responsible for accountability from receipt to return or destruction, and accountability records are reconciled at each monitoring visit.

5.3 Assignment to treatment

Subjects are assigned by an interactive response system after eligibility is confirmed. There is no randomisation; every eligible subject receives study treatment.

5.4 Blinding

This is an open-label study. Investigators and subjects are aware of the assigned treatment.

5.5 Treatment compliance

Administration is recorded for every dose, with the date, the dose given and the reason for any omission or change. Relative dose intensity is derived as the dose received divided by the dose planned over the same interval and is summarised in the clinical study report.

Compliance is assessed from the administration record rather than from subject report. Every dose is recorded with its date, the amount given and, where it differs from the amount planned, the reason.

Relative dose intensity is derived as the dose actually received divided by the dose planned over the same interval, and is reported by treatment group in the clinical study report. It is the measure that distinguishes a regimen that is tolerable from one that is only nominally tolerable because subjects are missing doses.

5.6 Dose modification

Modification for toxicity follows Table 4. Dose reduction is permitted to the levels in Table 3; re-escalation after a reduction is not permitted.

Table 4 Dose modification for toxicity

Event and grade	Action for the current dose	Action on recovery
Grade 1	Continue at the current dose.	No change.
Grade 2, first occurrence	Continue at the current dose with supportive care; consider interruption if the event does not settle within 7 days.	Resume at the same dose.
Grade 2, recurrent	Interrupt until recovery to grade 1 or baseline.	Resume at the same dose, or one dose level lower if the interruption exceeded 14 days.
Grade 3	Interrupt until recovery to grade 1 or baseline.	Resume at one dose level lower.
Grade 3, recurrent after one reduction	Interrupt until recovery to grade 1 or baseline.	Resume at a second reduced dose level, or discontinue.
Grade 4	Discontinue study treatment permanently.	Not applicable.
Myelodysplastic syndrome or acute myeloid leukaemia, any grade at or above 2	Interrupt and manage according to Appendix 5. Seek the opinion of the medical monitor before any further dose.	Resume at one dose level lower only if the event resolved to grade 1 or baseline within 28 days.
Pneumonitis, any grade at or above 2	Interrupt and manage according to Appendix 5. Seek the opinion of the medical monitor before any further dose.	Resume at one dose level lower only if the event resolved to grade 1 or baseline within 28 days.
Severe anaemia, grade 3 or above	Interrupt and manage according to Appendix 5. Seek the opinion of the medical monitor before any further dose.	Resume at one dose level lower only if the event resolved to grade 1 or baseline within 28 days.

Dose levels are defined in Section 5.1. A subject who requires more than two dose reductions discontinues study treatment. Interruptions longer than 28 days require the agreement of the medical monitor for the subject to continue.

5.6.1 Dosing schedule

Kelvaparib is supplied as film-coated tablets and administered orally twice daily on a continuous schedule. The schedule administers the assigned dose with weekly blood counts in weeks 1 to 4, the assigned dose with monthly blood counts in week 5 onwards.

5.6.2 Myelosuppression prophylaxis and monitoring

All subjects receive full blood count before treatment, weekly for the first month and monthly thereafter, with dose interruption for grade 3 or higher cytopenia and prompt haematology referral for any cytopenia persisting beyond four weeks. Subjects at high risk, defined as a haemoglobin below 110 g/L at baseline, three or more prior lines of platinum-based therapy, or prior pelvic radiotherapy, additionally receive intravenous hydration and laboratory monitoring before the first dose and at 4, 8, 12 and 24 hours after it.

5.6.3 Modification for myelosuppression

For grade 4 haematological toxicity or grade 3 or higher non-haematological toxicity considered related to study treatment, dosing is withheld until recovery to grade 1 or baseline and resumed at the next lower dose level. No more than two dose reductions are permitted. For laboratory myelosuppression, dosing is withheld with daily monitoring; treatment resumes at the same dose with intensified prophylaxis if laboratory-only, and at the next lower dose level if clinical myelosuppression occurred.

Anticancer therapy other than study treatment is prohibited. Strong CYP3A4 inhibitors and inducers are prohibited. Where a moderate inhibitor cannot be avoided the dose is reduced by at least 50 per cent and the subject is monitored under the high-risk schedule.

Full blood count before treatment, weekly for the first month and monthly thereafter, with dose interruption for grade 3 or higher cytopenia and prompt haematology referral for any cytopenia persisting beyond four weeks is required for every subject and is started before the first dose. Where a subject cannot receive it, the reason is recorded and the subject is monitored under the arrangements for high-risk subjects.

5.7 Concomitant therapy

All medication taken from 28 days before the first dose until the safety follow-up visit is recorded, with the indication, dose, route and dates. Supportive care is given according to local practice. Live attenuated vaccines, other investigational agents and other systemic anti-cancer therapy are prohibited during the treatment period. Medication known to prolong the corrected QT interval or to be a strong inhibitor or inducer of the principal clearance pathway should be avoided; where it cannot be, the interaction and the monitoring applied are recorded.

5.8 Continued access after the study

A subject still deriving benefit at the end of the study may continue to receive Kelvaparib under a separate protocol or, where no such protocol exists, through the sponsor's post-trial access arrangements, subject to regulatory and ethics approval in the territory concerned.

6 DISCONTINUATION AND WITHDRAWAL

6.1 Discontinuation of study treatment

- Documented disease progression, unless the investigator judges continued treatment to be in the subject's interest and the sponsor agrees.

- An adverse event meeting a permanent discontinuation criterion in Table 4.
- Pregnancy.
- A dose interruption longer than 28 days, unless the medical monitor agrees to continuation.
- Withdrawal of consent to treatment.
- Any change in condition that means the subject no longer meets the eligibility criteria in a way that affects safety.

A subject who discontinues study treatment attends the end of treatment visit and the safety follow-up visit, and remains in survival follow-up unless consent for that is also withdrawn.

Discontinuation of study treatment is not withdrawal from the study. A subject who stops treatment remains in follow-up under Section 6.2 unless consent for that is also withdrawn, because the information that matters most about a subject who stopped is often what happened afterwards.

The decision to stop treatment rests with the investigator in consultation with the subject. Where the reason is an adverse event, the event is followed to resolution or stabilisation and the outcome recorded, whether or not the subject remains in the study.

6.2 Withdrawal from the study

A subject may withdraw at any time without giving a reason and without prejudice to their care. Data collected before withdrawal are retained and analysed. The date of withdrawal and the reason, where volunteered, are recorded.

Treatment continues until the subject no longer derives benefit, until unacceptable toxicity, or until consent is withdrawn.

6.3 Lost to follow-up

A subject is lost to follow-up only after at least three documented attempts to make contact, including one by recorded delivery, and after the last known date of contact has been recorded. Public records are consulted for survival status where local law permits.

7 STUDY ASSESSMENTS AND PROCEDURES

Assessments are performed at the visits shown in Table 5. Where a visit window is specified, an assessment performed inside the window is not a deviation.

7.1 Efficacy assessments

Disease is assessed according to RECIST version 1.1 at screening and at the intervals in Table 5, by the same modality throughout for a given subject wherever possible. A response is confirmed by a repeat assessment no less than 4 weeks later. Assessments continue on schedule until progression is documented, irrespective of dose interruption.

The same modality is used throughout for a given subject wherever possible, because a change of modality introduces a change in measured extent that can be mistaken for a change in disease.

A response is confirmed by a repeat assessment no less than 4 weeks later. An unconfirmed response is recorded but is not counted as a response in the primary analysis, and the analysis plan states how it is handled in the sensitivity analyses.

Assessments continue on schedule after a dose interruption and after a dose reduction. They stop only when progression is documented, when the subject withdraws consent to further assessment, or at the end of the study.

7.1.1 Response assessment for this study

Response is assessed according to RECIST version 1.1 at the end of cycles 3, 6, 9 and 12 and every six cycles thereafter, comprising physical examination and cross-sectional imaging of the chest, abdomen and pelvis by computed tomography or magnetic resonance imaging, with the same modality used throughout for a given subject. A response must be confirmed by a repeat assessment no fewer than four weeks later.

7.1.2 Study-specific schedule of assessments

Table 5 Schedule of Assessments

Assessment	Screening	Monitoring wk 1-4	C1D1	Each cycle D1	End of treatment	Follow-up
Informed consent	X					
Eligibility criteria	X					
Medical history	X					
Physical examination	X	X	X	X	X	X
ECOG performance status	X		X	X	X	X
Vital signs	X	X	X	X	X	X
Body weight	X		X	X	X	
12-lead electrocardiogram	X	X	X	X	X	
Haematology	X	X	X	X	X	X
Clinical chemistry	X	X	X	X	X	X
Coagulation	X		X	X	X	
Urinalysis	X		X	X	X	
Myelosuppression laboratory panel	X	X	X			
Pregnancy test	X		X	X	X	
Computed tomography or magnetic resonance imaging	X			C3, C6, C9, C12, then every 6 cycles	X	X
Tumour response assessment				C3, C6, C9, C12, then every 6 cycles	X	X
Pharmacokinetic sampling		X	X	C2, C3		
Biomarker sampling	X		X	C2, C3, C6	X	
Adverse event recording		X	X	X	X	X
Concomitant medication	X	X	X	X	X	
Study drug accountability		X	X	X	X	
Survival status						X

C = cycle; D = day. Assessments during the first month of treatment are performed weekly. Subjects at high risk undergo additional laboratory monitoring before dosing and at 4, 8, 12 and 24 hours after the first dose.

7.2 Safety assessments

Safety assessment comprises adverse event enquiry, physical examination, vital signs, performance status, 12-lead electrocardiogram and the laboratory panels in Table 8 (Appendix 2). Laboratory results are graded using the Common Terminology Criteria for Adverse Events.

Safety assessment is deliberately more frequent early in treatment, when the events of special interest are most likely to occur, and is reduced once a subject has passed the period of greatest risk without incident. The schedule in Table 5 reflects that.

Laboratory results are graded using the Common Terminology Criteria for Adverse Events. A laboratory abnormality is recorded as an adverse event where it is clinically significant, where it requires intervention, or where it leads to a change in study treatment; otherwise it is captured as laboratory data alone. This convention is applied consistently, because inconsistency between sites in what is recorded as an event is a common source of apparent difference in safety profile.

Vital signs are measured after the subject has been at rest for at least five minutes and, where a post-dose measurement is required, at the interval specified for the route of administration. Electrocardiograms are recorded in triplicate at screening and as single tracings thereafter unless an abnormality requires repetition.

All samples are analysed at the central laboratory. Local results may be used for immediate clinical decisions and are recorded where they lead to a dose change.

Haematology. The expected haematological effect of this product is the reason for the frequency of this panel early in treatment. A fall that is progressive across consecutive measurements is treated differently from a single low value, and the dose modification rules in Table 4 are applied to the confirmed value.

Clinical chemistry. Hepatic and renal parameters are the ones that most often lead to a change in study treatment. A rise in transaminase together with a rise in bilirubin is handled under Appendix 10 rather than under the general rules, because the combination has a specific meaning.

Coagulation. Assessed at the intervals in Table 5 and whenever a bleeding event occurs. Results are interpreted alongside the platelet count rather than in isolation.

Urinalysis. A screening test. An abnormal result is followed by microscopy and, where it persists, by quantification of proteinuria.

Serology at screening. Performed once, to establish eligibility and to identify subjects who require monitoring for reactivation.

Pregnancy testing. Performed at screening and at the intervals in Table 5. A positive result requires immediate discontinuation and reporting under Appendix 4.

7.3 Adverse events and serious adverse events

Definitions, the period of collection, the assessment of severity and causality, and the reporting route and timelines are in Appendix 3. Adverse events of special interest for this product are: myelodysplastic syndrome or acute myeloid leukaemia; pneumonitis; severe anaemia. Each is reported to the sponsor within 24 hours of the site becoming aware, whether or not it meets a seriousness criterion.

7.3.1 Definition of dose-limiting toxicity

A dose-limiting toxicity is an adverse event of grade 3 or higher, considered at least possibly related to study treatment, with onset during the DLT observation period: the first 21 days of treatment at the assigned cohort dose, within the 28-day first cycle. A subject is evaluable for dose-limiting toxicity if they received at least 21 days of study treatment or had a dose-limiting toxicity.

Adverse events of special interest for this programme are myelodysplastic syndrome or acute myeloid leukaemia, pneumonitis, severe anaemia, and are reported within 24 hours irrespective of seriousness.

7.3.2 Events anticipated with this product

The events below are those anticipated in this population on the basis of the mechanism and of the clinical experience to date. Listing them here does not limit the reporting of any other event.

Nausea (Gastrointestinal disorders). Anticipated as very common in this population and is usually grade 1 or 2. It falls within gastrointestinal disorders. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Anaemia (Blood and lymphatic system disorders). Anticipated as very common in this population and frequently reaches grade 3 or above. It falls within blood and lymphatic system disorders. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Fatigue (General disorders and administration site conditions). Anticipated as very common in this population and occasionally reaches grade 3 or above. It falls within general disorders and administration site conditions. Manage under Table 4: continue at grade 1, withhold at grade 2

until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Vomiting (Gastrointestinal disorders). Anticipated as very common in this population and is usually grade 1 or 2. It falls within gastrointestinal disorders. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Constipation (Gastrointestinal disorders). Anticipated as very common in this population and is usually grade 1 or 2. It falls within gastrointestinal disorders. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Decreased appetite (Metabolism and nutrition disorders). Anticipated as very common in this population and is usually grade 1 or 2. It falls within metabolism and nutrition disorders. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Thrombocytopenia (Blood and lymphatic system disorders). Anticipated as very common in this population and occasionally reaches grade 3 or above. It falls within blood and lymphatic system disorders. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Neutropenia (Blood and lymphatic system disorders). Anticipated as very common in this population and occasionally reaches grade 3 or above. It falls within blood and lymphatic system disorders. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Headache (Nervous system disorders). Anticipated as very common in this population and is usually grade 1 or 2. It falls within nervous system disorders. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade

3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Diarrhoea (Gastrointestinal disorders). Anticipated as very common in this population and is usually grade 1 or 2. It falls within gastrointestinal disorders. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Dysgeusia (Nervous system disorders). Anticipated as very common in this population and is usually grade 1 or 2. It falls within nervous system disorders. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Abdominal pain (Gastrointestinal disorders). Anticipated as very common in this population and is usually grade 1 or 2. It falls within gastrointestinal disorders. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Asthenia (General disorders and administration site conditions). Anticipated as very common in this population and is usually grade 1 or 2. It falls within general disorders and administration site conditions. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Dizziness (Nervous system disorders). Anticipated as very common in this population and is usually grade 1 or 2. It falls within nervous system disorders. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Insomnia (Psychiatric disorders). Anticipated as very common in this population and is usually grade 1 or 2. It falls within psychiatric disorders. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and

discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Cough (Respiratory, thoracic and mediastinal disorders). Anticipated as very common in this population and is usually grade 1 or 2. It falls within respiratory, thoracic and mediastinal disorders. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Arthralgia (Musculoskeletal and connective tissue disorders). Anticipated as very common in this population and is usually grade 1 or 2. It falls within musculoskeletal and connective tissue disorders. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Nasopharyngitis (Infections and infestations). Anticipated as very common in this population and is usually grade 1 or 2. It falls within infections and infestations. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Dyspepsia (Gastrointestinal disorders). Anticipated as very common in this population and is usually grade 1 or 2. It falls within gastrointestinal disorders. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Back pain (Musculoskeletal and connective tissue disorders). Anticipated as very common in this population and is usually grade 1 or 2. It falls within musculoskeletal and connective tissue disorders. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Oedema peripheral (General disorders and administration site conditions). Anticipated as very common in this population and is usually grade 1 or 2. It falls within general disorders and administration site conditions. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade

4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Leukopenia (Blood and lymphatic system disorders). Anticipated as very common in this population and occasionally reaches grade 3 or above. It falls within blood and lymphatic system disorders. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Urinary tract infection (Infections and infestations). Anticipated as very common in this population and occasionally reaches grade 3 or above. It falls within infections and infestations. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Alanine aminotransferase increased (Investigations). Anticipated as common in this population and occasionally reaches grade 3 or above. It falls within investigations. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Dyspnoea (Respiratory, thoracic and mediastinal disorders). Anticipated as common in this population and occasionally reaches grade 3 or above. It falls within respiratory, thoracic and mediastinal disorders. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Weight decreased (Investigations). Anticipated as common in this population and is usually grade 1 or 2. It falls within investigations. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Lymphopenia (Blood and lymphatic system disorders). Anticipated as common in this population and frequently reaches grade 3 or above. It falls within blood and lymphatic system disorders. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or

baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Myalgia (Musculoskeletal and connective tissue disorders). Anticipated as common in this population and is usually grade 1 or 2. It falls within musculoskeletal and connective tissue disorders. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Aspartate aminotransferase increased (Investigations). Anticipated as common in this population and occasionally reaches grade 3 or above. It falls within investigations. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Pyrexia (General disorders and administration site conditions). Anticipated as common in this population and is usually grade 1 or 2. It falls within general disorders and administration site conditions. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Rash (Skin and subcutaneous tissue disorders). Anticipated as common in this population and is usually grade 1 or 2. It falls within skin and subcutaneous tissue disorders. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Hypertension (Vascular disorders). Anticipated as common in this population and occasionally reaches grade 3 or above. It falls within vascular disorders. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Blood creatinine increased (Investigations). Anticipated as common in this population and occasionally reaches grade 3 or above. It falls within investigations. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and

reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Pneumonia (Infections and infestations). Anticipated as common in this population and frequently reaches grade 3 or above. It falls within infections and infestations. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Pneumonitis (Respiratory, thoracic and mediastinal disorders). Anticipated as common in this population and frequently reaches grade 3 or above. It falls within respiratory, thoracic and mediastinal disorders. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered. This is an event of special interest and is reported to the sponsor within 24 hours whether or not it is serious.

Myelodysplastic syndrome (Neoplasms benign, malignant and unspecified). Anticipated as common in this population and frequently reaches grade 3 or above. It falls within neoplasms benign, malignant and unspecified. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

Sepsis (Infections and infestations). Anticipated as common in this population and frequently reaches grade 3 or above. It falls within infections and infestations. Manage under Table 4: continue at grade 1, withhold at grade 2 until recovery to grade 1 or baseline, withhold and reduce at grade 3, and discontinue at grade 4. Investigate for causes other than study treatment before attributing it, and record the alternative considered.

7.4 Pharmacokinetics

Blood samples for the measurement of Kelvaparib in plasma are taken at the times in Table 5. Samples are assayed by a validated method at a central bioanalytical facility. Parameters derived may include maximum observed concentration, time to that concentration, area under the concentration-time curve, apparent clearance, apparent volume of distribution, terminal half-life and accumulation ratio.

Sparse samples are collected pre-dose at the visits set out in the schedule of assessments. A subset undergoes serial sampling at pre-dose and 0.5, 1, 2, 4, 6, 8, 12 and 24 hours on day 8 of the assigned dose. Samples are analysed by a validated liquid chromatography tandem mass spectrometry method with a lower limit of quantification of 1.00 ng/mL.

7.5 Pharmacodynamics and biomarkers

Exploratory samples are taken at the times in Table 5 for candidate biomarkers that may relate to mechanism, response or resistance. These analyses are exploratory and are not used in any decision affecting the conduct of the study.

7.6 Immunogenicity

Kelvaparib is a small-molecule medicine and is not expected to provoke an antibody response; no immunogenicity samples are taken in this study.

7.7 Health status and resource use

Health status is recorded using the instrument specified in the protocol, completed by the subject before any other assessment at that visit so that the response is not influenced by the discussion of results.

7.8 Description of procedures by visit

This section describes what is done at each visit. Where it and Table 5 differ, Table 5 governs.

7.8.1 Screening (up to 28 days before the first dose)

Consent is taken before anything else is done. Eligibility is then established against every criterion in Sections 4.1 and 4.2, and the source of each determination is recorded. Demography, the history of the disease including every prior line of therapy with its dates and best response, concurrent conditions and current medication are collected. Physical examination, vital signs, body weight, height and performance status are recorded. A 12-lead electrocardiogram is taken in triplicate. The full laboratory panel in Table 8 (Appendix 2) is drawn, together with serology and, for women of childbearing potential, a serum pregnancy test. Disease is assessed by the imaging modality that will be used throughout for that subject. Where a screening assessment falls outside the window because of scheduling, the subject may be re-screened once.

7.8.2 Cycle 1 Day 1 (day 1)

Eligibility is confirmed and the treatment assignment obtained from the interactive response system. Pre-dose assessments comprise physical examination, vital signs, performance status, electrocardiogram, the laboratory panel and, where required, a pregnancy test. Pre-dose samples for pharmacokinetics and biomarkers are drawn before the first administration. The first dose is given under direct observation. Vital signs are repeated after administration at the interval specified for the route used. Adverse events and concomitant medication are reviewed before the subject leaves.

7.8.3 Cycle 1 Day 8 and Day 15 (days 8 and 15)

Vital signs, the laboratory panel, adverse event review and concomitant medication review are performed before dosing. Pharmacokinetic samples are drawn at the times in Table 5. The dose is given if the laboratory results meet the criteria for administration in Table 4; where they do not, the dose is withheld and the reason recorded.

7.8.4 Cycle 2 Day 1 (day 29)

The assessments of a treatment visit are repeated in full, including physical examination, performance status, electrocardiogram and the complete laboratory panel. Biomarker samples are drawn. Relative dose intensity for cycle 1 is reconciled against the administration record.

7.8.5 Cycle 3 Day 1 and subsequent cycles (day 57 onwards)

Treatment visits continue on the schedule in Table 5. The first on-treatment disease assessment is performed at the end of cycle 3 and thereafter at the interval in Table 5, by the modality used at screening. A response is confirmed by a repeat assessment no less than 4 weeks later. Disease assessment continues on its own schedule irrespective of dose delay.

7.8.6 End of treatment (within 7 days of the decision to stop)

The reason for discontinuation is recorded against the categories in Section 6.1. Physical examination, vital signs, performance status, electrocardiogram and the full laboratory panel are performed. A final disease assessment is performed unless one was performed within the preceding 4 weeks. Unused study treatment is returned and reconciled.

7.8.7 Safety follow-up (30 days after the last dose)

Adverse events are reviewed and any event ongoing at the end of treatment is followed to resolution, stabilisation or the point at which no further follow-up is possible. The laboratory panel is repeated. A pregnancy test is performed for women of childbearing potential. Any new anti-disease therapy started since the last dose is recorded with its start date.

7.8.8 Survival follow-up (every 12 weeks)

Contact is made by clinic visit, telephone or review of the medical record to establish survival status and any subsequent anti-disease therapy. Follow-up continues until death, withdrawal of consent to follow-up, loss to follow-up as defined in Section 6.3, or the end of the study.

7.9 Conduct of each procedure

How each procedure is performed, and what is recorded, is given below. Consistency between sites in how a procedure is carried out matters as much as whether it is carried out at all.

Informed consent. Taken by a person authorised by the investigator, after adequate time and in a language the subject understands. The version taken is recorded, and consent is retaken if the information sheet is revised materially.

Eligibility criteria. Each criterion is checked against source and the source of each determination is recorded. Where a value is borderline the assessment is repeated once within the screening window rather than accepted on a single measurement.

Demography and medical history. History covers concurrent conditions, prior therapy for the disease under study with dates and best response, and any condition relevant to the eligibility criteria. It is recorded once and updated only where new information about the past emerges.

Physical examination. A full examination at screening and at the end of treatment; a symptom-directed examination at other visits. A new finding is recorded as an adverse event unless it was present at baseline and has not changed.

Vital signs. Taken after at least five minutes at rest, in the same posture each time for a given subject. Where the route requires post-dose observation, they are repeated at the specified interval.

Performance status. Recorded by the same scale throughout and, where practicable, by the same assessor, because the scale is coarse and inter-assessor variation is the largest source of apparent change.

Body weight. Recorded at the visits in Table 5 as part of the safety assessment. The dose of Kelvaparib is fixed and does not depend on body weight.

12-lead electrocardiogram. In triplicate at screening and as a single tracing thereafter, taken after at least five minutes supine and before dosing where both fall on the same day. An abnormal tracing is repeated and reviewed by a clinician before the next dose.

Haematology. Drawn before dosing at every visit where dosing occurs, so that the result is available to the decision to dose. Where a local result is used for that decision the central sample is still drawn.

Clinical chemistry. As for haematology. Hepatic parameters are reviewed against Appendix 10 at every visit rather than only when an abnormality is flagged.

Coagulation. At the visits in Table 5 and whenever a bleeding event occurs or an anticoagulant is started or changed.

Urinalysis. By dipstick at the visits in Table 5, with microscopy and quantification where any result is abnormal.

Pregnancy test. Serum at screening; urine or serum thereafter at the intervals in Table 5 and at the end of treatment. A positive result stops study treatment immediately.

Study treatment administration. Given only after the pre-dose assessments have been reviewed. The dose given, the time, and any deviation from the planned dose with its reason are recorded at the time of administration rather than retrospectively.

Pharmacokinetic sampling. Taken at the nominal times in Table 5 with the actual time recorded to the minute. A sample taken outside its window is still analysed and is flagged, because the actual time is what the analysis uses.

Biomarker sampling. Exploratory. Omission does not constitute a deviation affecting the primary endpoint, and a subject may decline it under the consent arrangements in Appendix 16.

Disease assessment. By the modality used at screening, on the calendar schedule in Table 5 irrespective of dose delay. Read locally for clinical decisions and, where the protocol requires it, centrally for the primary analysis.

Adverse event review. By open question before any directed enquiry, so that the subject's own account is captured before it is shaped by the question. Every event is graded and assessed for causality at the time it is recorded.

Concomitant medication review. Covers prescribed and non-prescribed medication, including herbal and over-the-counter preparations, with the indication for each. A new medication started for an adverse event is linked to that event.

Survival status. Established by visit, telephone or record review. Where contact fails, the attempts are documented before a subject is classified as lost to follow-up under Section 6.3.

8 STATISTICAL CONSIDERATIONS

8.1 Sample size determination

No formal hypothesis testing is planned. Sample size follows the operating characteristics of the 3+3 design and the number of subjects required to characterise safety at the selected dose. Proportions are presented with exact Clopper-Pearson 95 per cent confidence intervals and time-to-event endpoints estimated by the Kaplan-Meier method.

The sample size is chosen so that the primary estimate is precise enough to support the conclusion the study is designed to reach, not merely so that a difference can be declared. The assumptions are stated so that a reader can see what the study would and would not detect.

The assumptions are drawn from the outcomes reported for this population and from the clinical experience with the product to date. They are assumptions, not predictions, and the study is not invalidated if they prove wrong; the effect of their being wrong on the precision of the estimate is examined in the analysis plan.

8.2 Analysis populations

Table 6 Analysis populations

Population	Definition
Screened	All subjects who gave informed consent.
Enrolled	All subjects assigned to treatment.
Safety	All subjects who received at least one dose of study treatment, analysed as treated.
DLT evaluable	Subjects who received at least 21 days of study treatment or had a dose-limiting toxicity.
Full analysis	All subjects in the safety analysis set with at least one post-baseline disease assessment, analysed as assigned.
Efficacy evaluable	The same subjects as the full analysis set.
Per protocol	Subjects in the full analysis set without a protocol deviation leading to discontinuation of study treatment.
Pharmacokinetic	Subjects with at least one measurable post-dose concentration and no protocol deviation affecting the interpretation of that concentration.

Screened analysis set. Used for the accounting of who entered the study and why they did not proceed. It is the denominator for the screen failure rate, which is the measure of whether the eligibility criteria are workable.

Enrolled analysis set. Used for disposition and for deviations. A subject assigned to treatment belongs to the study whether or not they went on to receive it.

Safety analysis set. Analysed as treated rather than as assigned, because an adverse event belongs to the treatment the subject actually received. This is the only population for which that convention applies.

Full analysis set. The population of every efficacy analysis: all subjects in the safety analysis set with at least one post-baseline disease assessment, analysed as assigned.

Per protocol analysis set. Used only for sensitivity analysis. It is not the primary population because membership of it depends on events after assignment, which can differ between treatment groups.

Efficacy evaluable analysis set. The same subjects as the full analysis set (at least one post-baseline disease assessment); the name is kept because the response tables use it.

Pharmacokinetic analysis set. Restricted to subjects whose sampling and dosing records support interpretation of the concentration, because a concentration whose timing relative to the dose is uncertain is not informative.

8.3 Primary analysis

Incidence of dose-limiting toxicity during the first 21 days at the assigned cohort dose, and the maximum tolerated dose derived from it.

8.4 Secondary analyses

Secondary endpoints are summarised descriptively by treatment group. Time-to-event endpoints are estimated by the Kaplan-Meier method with the median and its confidence interval; binary endpoints are presented as proportions with exact confidence intervals.

8.5 Multiplicity

No multiplicity adjustment is required because only one primary comparison is performed and all secondary analyses are descriptive. Nominal p-values, where presented, are not adjusted and are not used to claim significance.

The safety analysis set comprises every subject who received at least one dose of study treatment, whether or not that dose was complete, and subjects are analysed according to the treatment actually received.

8.6 Interim analysis

Any interim analysis, its timing and the quantities examined are prespecified in the statistical analysis plan, and are finalised before the first interim data are examined. No alpha is spent at an interim analysis.

8.7 Handling of missing data

Missing data are not imputed for the primary analysis. A subject without the event of interest is censored at the last assessment at which the absence of the event was documented. The extent and pattern of missingness are reported, and the sensitivity analyses in the statistical analysis plan examine whether the conclusion depends on the censoring rule.

The extent of missing data, and the reasons for it, are reported for every variable contributing to the primary endpoint. Missingness that is related to outcome is the problem; missingness that is not is a loss of precision only, and the two are distinguished as far as the data allow.

The primary analysis does not impute. The sensitivity analyses in the statistical analysis plan examine whether the conclusion depends on the censoring rule, including an analysis in which subjects with no post-baseline assessment are treated as failures and one in which subsequent therapy is ignored.

8.8 Changes to the planned analysis

Any change to the planned analysis made after the protocol is approved is documented in the statistical analysis plan before database lock. A change made after lock is reported as such in the clinical study report, with the reason and its effect on the conclusion.

9 ETHICS AND STUDY ADMINISTRATION

9.1 Regulatory and ethical considerations

The study is conducted in accordance with the protocol, the principles of the Declaration of Helsinki, the principles of Good Clinical Practice and the regulatory requirements of each territory. The protocol, the subject information sheet and the consent form are approved by the responsible ethics committee before any subject is enrolled at that site.

9.1.1 Independent data monitoring committee

No independent data monitoring committee is appointed for this study. Accumulating safety data are reviewed by the sponsor under the safety monitoring plan.

9.1.2 Data handling and quality

Data are recorded on an electronic case report form and managed by Steinbeck Clinical Research GmbH. Source data verification covers informed consent, eligibility, dose administration, serious adverse events and primary endpoint data for all subjects. Essential documents are retained for at least 25 years.

9.2 Informed consent

Consent is obtained in writing before any study-specific procedure, by a person authorised by the investigator, after adequate time for consideration and for questions. The subject receives a copy of the signed form. Consent is re-obtained if the information sheet is revised in a way that could affect the subject's willingness to continue.

9.3 Confidentiality and data protection

Subjects are identified in study records by number, never by name. Personal data are processed in accordance with the applicable data protection law of the territory. The sponsor, the responsible authorities and the ethics committee may inspect the source records.

9.4 Data handling and record keeping

Data are recorded in the electronic case report form by trained site staff and are subject to programmed and manual checks. Source documents remain at the site. Essential documents are retained for the period required in each territory and for at least 25 years, and are not destroyed without the written agreement of the sponsor.

9.5 Monitoring, audit and inspection

Monitoring is risk-based and described in the monitoring plan. The monitor verifies consent, eligibility, the reporting of serious adverse events and the primary endpoint against source for every subject. Sites may be audited by the sponsor's quality assurance function and inspected by the authorities.

9.6 Protocol deviations

A deviation is any departure from the approved protocol. Deviations are recorded, categorised as important or not by prespecified criteria before database lock, and reported in the clinical study report. A deviation is not permitted in advance except where necessary to eliminate an immediate hazard to a subject.

Deviations are recorded as they are identified rather than reconstructed at the end of the study, and the criteria that make a deviation important are agreed and documented before the database is locked, so that the classification cannot be influenced by knowledge of the result.

A deviation is not authorised in advance. Where a departure from the protocol is necessary to eliminate an immediate hazard to a subject, it is made and then reported to the sponsor and the ethics committee; it remains a deviation and is recorded as one.

9.7 Publication policy

The results of the study are published irrespective of outcome, and are registered and reported on the public registers required in each territory within the applicable timelines.

9.8 Protocol amendments

An amendment is implemented only after approval by the responsible authorities and ethics committees, except where it is necessary to eliminate an immediate hazard. The history of amendments to this protocol is in Appendix 6.

9.9 Insurance and indemnity

The sponsor holds insurance covering its liabilities arising from the conduct of this study in each territory in which the study is conducted, and indemnifies investigators and institutions against claims arising from the proper conduct of the study in accordance with the protocol. Evidence of cover is provided to each ethics committee that requires it.

9.10 Financing and conflicts of interest

The study is funded by the sponsor. Financial arrangements with each institution are set out in a separate clinical trial agreement. Investigators disclose financial interests to the sponsor before the study begins and whenever an interest arises or changes during the study, and those disclosures are retained for the period required by the applicable regulations.

9.11 Compensation and reimbursement of subjects

Subjects are not paid to take part. Reasonable expenses of travel and subsistence directly attributable to participation are reimbursed where local practice and the ethics committee permit, and the arrangements are described in the subject information sheet.

9.12 Termination of the study or of a site

The sponsor may terminate the study or the participation of a site at any time. Where termination is for reasons of safety, the sponsor notifies investigators, ethics committees and the responsible authorities immediately and provides for the continuing care of subjects already enrolled. Where a site is closed for reasons of conduct, the sponsor arranges for subjects to be transferred to another site where that is practicable.

10 PROTOCOL AMENDMENT SUMMARY

Amendment	Date	Principal change
1	1 September 2017	Clarification of eligibility and monitoring

11 REFERENCES

References cited in this protocol are listed in the reference section of the current investigator brochure and in the bibliography maintained with the protocol file.

- Eisenhauer EA, et al. New response evaluation criteria in solid tumours: revised RECIST guideline version 1.1. Eur J Cancer. 2009;45(2):228-247.

- Common Terminology Criteria for Adverse Events version 5.0. National Cancer Institute; 2017.
- International Council for Harmonisation. Guideline for Good Clinical Practice E6(R2). 2016.

Appendix 1 ABBREVIATIONS

Table 7 Abbreviations used in this protocol

Abbreviation	Meaning
AE	adverse event
ALT	alanine aminotransferase
AST	aspartate aminotransferase
AUC	area under the concentration-time curve
CI	confidence interval
CTCAE	Common Terminology Criteria for Adverse Events
ECG	electrocardiogram
ECOG	Eastern Cooperative Oncology Group
eCRF	electronic case report form
GCP	Good Clinical Practice
IEC	independent ethics committee
IRB	institutional review board
MedDRA	Medical Dictionary for Regulatory Activities
PK	pharmacokinetic
SAE	serious adverse event
SAP	statistical analysis plan
ULN	upper limit of normal

Appendix 2 CLINICAL LABORATORY TESTS

Table 8 Clinical laboratory assessments

Panel	Analytes
Haematology	Haemoglobin, haematocrit, red blood cell count, white blood cell count with differential, absolute neutrophil count, absolute lymphocyte count, platelet count, reticulocyte count
Clinical chemistry	Sodium, potassium, chloride, bicarbonate, urea, creatinine, estimated glomerular filtration rate, glucose, calcium, magnesium, phosphate, total protein, albumin, total and direct bilirubin, alanine aminotransferase, aspartate aminotransferase, alkaline phosphatase, gamma-glutamyl transferase, lactate dehydrogenase, urate
Coagulation	Prothrombin time, international normalised ratio, activated partial thromboplastin time, fibrinogen
Urinalysis	Specific gravity, pH, protein, glucose, ketones, blood, with microscopy if any result is abnormal
Serology at screening	Hepatitis B surface antigen, hepatitis B core antibody, hepatitis C antibody, human immunodeficiency virus antibody
Pregnancy testing	Serum beta human chorionic gonadotrophin at screening; urine or serum thereafter

All samples are analysed at the central laboratory. Local results may be used for immediate clinical decisions and are recorded where they lead to a dose change.

Appendix 3 ADVERSE EVENT DEFINITIONS AND REPORTING

An adverse event is any untoward medical occurrence in a subject administered a medicinal product, whether or not related to that product. A serious adverse event is one that results in death, is life-threatening, requires or prolongs hospitalisation, results in persistent or significant disability or incapacity, is a congenital anomaly or birth defect, or is an important medical event that may jeopardise the subject or require intervention to prevent one of the preceding outcomes.

Adverse events are collected from the signing of consent to the end of study participation, and serious adverse events considered related to study treatment are collected without time limit. Severity is graded using the Common Terminology Criteria for Adverse Events. Causality is assessed by the investigator as related or not related, and an assessment of not related requires an alternative explanation to be recorded.

Serious adverse events are reported to the sponsor within 24 hours of the site becoming aware. The sponsor assesses expectedness against the reference safety information in the investigator brochure and submits expedited reports to the authorities, ethics committees and investigators within the timelines required in each territory: 7 calendar days for a fatal or life-threatening suspected unexpected serious adverse reaction and 15 calendar days otherwise.

Appendix 4 CONTRACEPTION AND PREGNANCY

A woman is of childbearing potential unless she is postmenopausal, defined as 12 months of spontaneous amenorrhoea without an alternative cause, or has undergone documented permanent sterilisation. Highly effective methods are those with a failure rate below 1 per cent a year when used consistently and correctly, and include combined or progestogen-only hormonal contraception associated with inhibition of ovulation, an intrauterine device or hormone-releasing system, bilateral tubal occlusion, vasectomised partner and sexual abstinence where that is the preferred and usual lifestyle of the subject.

Contraception is required from screening until 90 days after the last dose for women and until 120 days after the last dose for men. A pregnancy occurring in a subject or in the partner of a male subject is reported to the sponsor within 24 hours and followed to outcome.

Appendix 5 MANAGEMENT OF EVENTS OF SPECIAL INTEREST

Myelodysplastic syndrome or acute myeloid leukaemia

Presentation. Myelodysplastic syndrome or acute myeloid leukaemia associated with Kelvaparib is most likely to occur during the period of most intensive monitoring in Table 5. Its early manifestations may be non-specific, which is why the monitoring is scheduled rather than prompted by symptoms.

Monitoring. The assessments in Table 5 are performed at the scheduled times irrespective of whether the subject reports symptoms. Any assessment suggesting the event is repeated within 24 to 72 hours, and the subject is reviewed by a clinician before the next dose.

Management at grade 1. Continue study treatment. Increase the frequency of the relevant assessment to at least weekly until the abnormality resolves or is stable, and record the event.

Management at grade 2. Withhold study treatment. Investigate for alternative causes, including concomitant medication and the underlying disease. Treat according to local practice and the guidance in the investigator brochure. Resume at the same dose only if the event resolves to grade 1 or baseline within 14 days; otherwise resume at one dose level lower.

Management at grade 3 or above. Withhold study treatment and involve the relevant specialist. Report to the sponsor within 24 hours whether or not a seriousness criterion is met. Rechallenge is permitted only after recovery to grade 1 or baseline within 28 days, at one dose level lower,

and with the written agreement of the medical monitor. Recurrence at grade 3 or above requires permanent discontinuation.

Reporting. Every occurrence is reported to the sponsor within 24 hours under Section 7.3. This applies whether or not the event is serious, so that a change in the frequency or severity of the event is detected across the study rather than at one site.

Pneumonitis

Presentation. Pneumonitis associated with Kelvaparib is most likely to occur during the period of most intensive monitoring in Table 5. Its early manifestations may be non-specific, which is why the monitoring is scheduled rather than prompted by symptoms.

Monitoring. The assessments in Table 5 are performed at the scheduled times irrespective of whether the subject reports symptoms. Any assessment suggesting the event is repeated within 24 to 72 hours, and the subject is reviewed by a clinician before the next dose.

Management at grade 1. Continue study treatment. Increase the frequency of the relevant assessment to at least weekly until the abnormality resolves or is stable, and record the event.

Management at grade 2. Withhold study treatment. Investigate for alternative causes, including concomitant medication and the underlying disease. Treat according to local practice and the guidance in the investigator brochure. Resume at the same dose only if the event resolves to grade 1 or baseline within 14 days; otherwise resume at one dose level lower.

Management at grade 3 or above. Withhold study treatment and involve the relevant specialist. Report to the sponsor within 24 hours whether or not a seriousness criterion is met. Rechallenge is permitted only after recovery to grade 1 or baseline within 28 days, at one dose level lower, and with the written agreement of the medical monitor. Recurrence at grade 3 or above requires permanent discontinuation.

Reporting. Every occurrence is reported to the sponsor within 24 hours under Section 7.3. This applies whether or not the event is serious, so that a change in the frequency or severity of the event is detected across the study rather than at one site.

Severe anaemia

Presentation. Severe anaemia associated with Kelvaparib is most likely to occur during the period of most intensive monitoring in Table 5. Its early manifestations may be non-specific, which is why the monitoring is scheduled rather than prompted by symptoms.

Monitoring. The assessments in Table 5 are performed at the scheduled times irrespective of whether the subject reports symptoms. Any assessment suggesting the event is repeated within 24 to 72 hours, and the subject is reviewed by a clinician before the next dose.

Management at grade 1. Continue study treatment. Increase the frequency of the relevant assessment to at least weekly until the abnormality resolves or is stable, and record the event.

Management at grade 2. Withhold study treatment. Investigate for alternative causes, including concomitant medication and the underlying disease. Treat according to local practice and the guidance in the investigator brochure. Resume at the same dose only if the event resolves to grade 1 or baseline within 14 days; otherwise resume at one dose level lower.

Management at grade 3 or above. Withhold study treatment and involve the relevant specialist. Report to the sponsor within 24 hours whether or not a seriousness criterion is met. Rechallenge is permitted only after recovery to grade 1 or baseline within 28 days, at one dose level lower, and with the written agreement of the medical monitor. Recurrence at grade 3 or above requires permanent discontinuation.

Reporting. Every occurrence is reported to the sponsor within 24 hours under Section 7.3. This applies whether or not the event is serious, so that a change in the frequency or severity of the event is detected across the study rather than at one site.

Appendix 6 PROTOCOL AMENDMENT HISTORY

Table 9 Protocol amendment history

Version	Date	Summary of change
1.0	17 October 2016	Original approved version.
2.0	1 September 2017	Clarification of eligibility and monitoring

Appendix 7 SEVERITY GRADING

Severity is graded using the Common Terminology Criteria for Adverse Events version 5.0. Where a term has no specific criterion in that publication, the general definitions below apply. Severity and seriousness are assessed separately: a severe event is not necessarily serious, and a serious event is not necessarily severe.

Table 10 General severity grading

Grade	Definition
Grade 1	Mild. Asymptomatic or mild symptoms; clinical or diagnostic observation only; intervention not indicated.
Grade 2	Moderate. Minimal, local or non-invasive intervention indicated; limiting age-appropriate instrumental activities of daily living.
Grade 3	Severe or medically significant but not immediately life-threatening; hospitalisation or prolongation of hospitalisation indicated; disabling; limiting self-care activities of daily living.
Grade 4	Life-threatening consequences; urgent intervention indicated.
Grade 5	Death related to the adverse event.

Table 11 Grading and required action for the events anticipated with this product

Preferred term	System organ class	Grade 1 to 2	Grade 3	Grade 4
Nausea	Gastrointestinal disorders	Asymptomatic or mild; observation or minimal intervention.	Severe; intervention or hospitalisation indicated; study treatment withheld per Table 4.	Life-threatening; urgent intervention; study treatment discontinued per Table 4.
Anaemia	Blood and lymphatic system disorders	Asymptomatic or mild; observation or minimal intervention.	Severe; intervention or hospitalisation indicated; study treatment withheld per Table 4.	Life-threatening; urgent intervention; study treatment discontinued per Table 4.
Fatigue	General disorders and administration site conditions	Asymptomatic or mild; observation or minimal intervention.	Severe; intervention or hospitalisation indicated; study treatment withheld per Table 4.	Life-threatening; urgent intervention; study treatment discontinued per Table 4.
Vomiting	Gastrointestinal disorders	Asymptomatic or mild; observation or minimal intervention.	Severe; intervention or hospitalisation indicated; study treatment withheld per Table 4.	Life-threatening; urgent intervention; study treatment discontinued per Table 4.
Constipation	Gastrointestinal disorders	Asymptomatic or mild; observation or minimal intervention.	Severe; intervention or hospitalisation indicated; study treatment withheld per Table 4.	Life-threatening; urgent intervention; study treatment discontinued per Table 4.
Decreased appetite	Metabolism and nutrition disorders	Asymptomatic or mild; observation or minimal intervention.	Severe; intervention or hospitalisation indicated; study treatment withheld per Table 4.	Life-threatening; urgent intervention; study treatment discontinued per Table 4.
Thrombocytopenia	Blood and lymphatic system disorders	Asymptomatic or mild; observation or minimal intervention.	Severe; intervention or hospitalisation indicated; study treatment withheld per Table 4.	Life-threatening; urgent intervention; study treatment discontinued per Table 4.

Preferred term	System organ class	Grade 1 to 2	Grade 3	Grade 4
Neutropenia	Blood and lymphatic system disorders	Asymptomatic or mild; observation or minimal intervention.	Severe; intervention or hospitalisation indicated; study treatment withheld per Table 4.	Life-threatening; urgent intervention; study treatment discontinued per Table 4.
Headache	Nervous system disorders	Asymptomatic or mild; observation or minimal intervention.	Severe; intervention or hospitalisation indicated; study treatment withheld per Table 4.	Life-threatening; urgent intervention; study treatment discontinued per Table 4.
Diarrhoea	Gastrointestinal disorders	Asymptomatic or mild; observation or minimal intervention.	Severe; intervention or hospitalisation indicated; study treatment withheld per Table 4.	Life-threatening; urgent intervention; study treatment discontinued per Table 4.
Dysgeusia	Nervous system disorders	Asymptomatic or mild; observation or minimal intervention.	Severe; intervention or hospitalisation indicated; study treatment withheld per Table 4.	Life-threatening; urgent intervention; study treatment discontinued per Table 4.
Abdominal pain	Gastrointestinal disorders	Asymptomatic or mild; observation or minimal intervention.	Severe; intervention or hospitalisation indicated; study treatment withheld per Table 4.	Life-threatening; urgent intervention; study treatment discontinued per Table 4.
Asthenia	General disorders and administration site conditions	Asymptomatic or mild; observation or minimal intervention.	Severe; intervention or hospitalisation indicated; study treatment withheld per Table 4.	Life-threatening; urgent intervention; study treatment discontinued per Table 4.
Dizziness	Nervous system disorders	Asymptomatic or mild; observation or minimal intervention.	Severe; intervention or hospitalisation indicated; study treatment withheld per Table 4.	Life-threatening; urgent intervention; study treatment discontinued per Table 4.

Preferred term	System organ class	Grade 1 to 2	Grade 3	Grade 4
Insomnia	Psychiatric disorders	Asymptomatic or mild; observation or minimal intervention.	Severe; intervention or hospitalisation indicated; study treatment withheld per Table 4.	Life-threatening; urgent intervention; study treatment discontinued per Table 4.
Cough	Respiratory, thoracic and mediastinal disorders	Asymptomatic or mild; observation or minimal intervention.	Severe; intervention or hospitalisation indicated; study treatment withheld per Table 4.	Life-threatening; urgent intervention; study treatment discontinued per Table 4.
Arthralgia	Musculoskeletal and connective tissue disorders	Asymptomatic or mild; observation or minimal intervention.	Severe; intervention or hospitalisation indicated; study treatment withheld per Table 4.	Life-threatening; urgent intervention; study treatment discontinued per Table 4.
Nasopharyngitis	Infections and infestations	Asymptomatic or mild; observation or minimal intervention.	Severe; intervention or hospitalisation indicated; study treatment withheld per Table 4.	Life-threatening; urgent intervention; study treatment discontinued per Table 4.
Dyspepsia	Gastrointestinal disorders	Asymptomatic or mild; observation or minimal intervention.	Severe; intervention or hospitalisation indicated; study treatment withheld per Table 4.	Life-threatening; urgent intervention; study treatment discontinued per Table 4.
Back pain	Musculoskeletal and connective tissue disorders	Asymptomatic or mild; observation or minimal intervention.	Severe; intervention or hospitalisation indicated; study treatment withheld per Table 4.	Life-threatening; urgent intervention; study treatment discontinued per Table 4.
Oedema peripheral	General disorders and administration site conditions	Asymptomatic or mild; observation or minimal intervention.	Severe; intervention or hospitalisation indicated; study treatment withheld per Table 4.	Life-threatening; urgent intervention; study treatment discontinued per Table 4.

Preferred term	System organ class	Grade 1 to 2	Grade 3	Grade 4
Leukopenia	Blood and lymphatic system disorders	Asymptomatic or mild; observation or minimal intervention.	Severe; intervention or hospitalisation indicated; study treatment withheld per Table 4.	Life-threatening; urgent intervention; study treatment discontinued per Table 4.
Urinary tract infection	Infections and infestations	Asymptomatic or mild; observation or minimal intervention.	Severe; intervention or hospitalisation indicated; study treatment withheld per Table 4.	Life-threatening; urgent intervention; study treatment discontinued per Table 4.
Alanine aminotransferase increased	Investigations	Asymptomatic or mild; observation or minimal intervention.	Severe; intervention or hospitalisation indicated; study treatment withheld per Table 4.	Life-threatening; urgent intervention; study treatment discontinued per Table 4.
Dyspnoea	Respiratory, thoracic and mediastinal disorders	Asymptomatic or mild; observation or minimal intervention.	Severe; intervention or hospitalisation indicated; study treatment withheld per Table 4.	Life-threatening; urgent intervention; study treatment discontinued per Table 4.
Weight decreased	Investigations	Asymptomatic or mild; observation or minimal intervention.	Severe; intervention or hospitalisation indicated; study treatment withheld per Table 4.	Life-threatening; urgent intervention; study treatment discontinued per Table 4.

This table lists the events for which specific guidance is given. It does not limit the reporting of any other event.

Appendix 8 PERFORMANCE STATUS SCALE

Table 12 Eastern Cooperative Oncology Group performance status

Score	Definition
0	Fully active, able to carry on all pre-disease performance without restriction.
1	Restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature.
2	Ambulatory and capable of all self-care but unable to carry out any work activities; up and about more than 50 per cent of waking hours.
3	Capable of only limited self-care; confined to bed or chair more than 50 per cent of waking hours.
4	Completely disabled; cannot carry on any self-care; totally confined to bed or chair.
5	Dead.

Appendix 9 LABORATORY REFERENCE RANGES

Reference ranges are those of the central laboratory and are supplied with every result. The ranges below are indicative and are used for the purposes of this protocol where a local range is not available.

Table 13 Indicative laboratory reference ranges

Test	Code	Unit	Lower	Upper	Panel
Alanine aminotransferase	ALT	U/L	7	45	Chemistry
Aspartate aminotransferase	AST	U/L	8	40	Chemistry
Albumin	ALB	g/L	35	50	Chemistry
Alkaline Phosphatase	ALP	U/L	40	130	Chemistry
Alpha-fetoprotein	AFP	ug/L	0	10	Chemistry
Bilirubin	BILI	umol/L	3	21	Chemistry
C-Reactive Protein	CRP	mg/L	0	5	Chemistry
Cancer Antigen 125	CA125	kU/L	0	35	Chemistry
Cancer Antigen 19-9	CA199	kU/L	0	37	Chemistry
Calcium	CA	mmol/L	2.2	2.6	Chemistry
Creatine Kinase	CK	U/L	30	200	Chemistry
Creatinine	CREAT	umol/L	50	100	Chemistry
Glucose	GLUC	mmol/L	3.9	5.8	Chemistry
Haemoglobin	HGB	g/L	120	160	Haematology
Haptoglobin	HAPTO	g/L	0.3	2	Chemistry
Troponin T, High Sensitivity	TROPONT	ng/L	0	14	Chemistry
International Normalised Ratio	INR	ratio	0.8	1.2	Coagulation
Immunoglobulin M	IGM	g/L	0.4	2.3	Chemistry
Lactate dehydrogenase	LDH	U/L	120	250	Chemistry
Leukocytes	WBC	10^9/L	4	11	Haematology
Lipase	LIPASE	U/L	13	60	Chemistry
Lymphocytes	LYM	10^9/L	1	4	Haematology
Magnesium	MG	mmol/L	0.7	1	Chemistry
N-Terminal Prohormone Brain Natriuretic Peptide	NTPROBNP	ng/L	0	125	Chemistry
Neutrophils	NEUT	10^9/L	2	7.5	Haematology
Phosphate	PHOS	mmol/L	0.8	1.5	Chemistry
Platelets	PLAT	10^9/L	150	400	Haematology
Potassium	K	mmol/L	3.5	5.1	Chemistry
Reticulocytes	RETI	10^9/L	25	100	Haematology
Immunoglobulin G	IGG	g/L	7	16	Chemistry
Immunoglobulin M	IGM	g/L	0.4	2.3	Chemistry
Sodium	NA	mmol/L	135	145	Chemistry
Immunoglobulin G, Total	IGGTOT	g/L	7	16	Chemistry
Bilirubin	BILI	umol/L	3	21	Chemistry
Urate	URATE	umol/L	150	420	Chemistry
Uric Acid	URATE	umol/L	150	420	Chemistry

Appendix 10 MANAGEMENT OF HEPATIC INJURY

An increase in alanine or aspartate aminotransferase above 3 times the upper limit of normal together with total bilirubin above 2 times the upper limit of normal, in the absence of cholestasis and with no alternative explanation, is a potential case of drug-induced liver injury and is reported to the sponsor within 24 hours whether or not it meets a seriousness criterion. Study treatment is withheld immediately. The subject is assessed for viral hepatitis, autoimmune hepatitis, biliary obstruction, alcohol, hypotension and concomitant hepatotoxic medication. Liver tests are repeated within 48 to 72 hours and then at least weekly until they return to baseline or stabilise. Rechallenge is permitted only with the written agreement of the sponsor's medical monitor.

Appendix 11 MANAGEMENT OF INFUSION-RELATED AND HYPERSENSITIVITY REACTIONS

Where study treatment is given by infusion, subjects are observed throughout and for the period after the infusion specified in the pharmacy manual. A mild reaction is managed by slowing the infusion and giving symptomatic treatment. A moderate reaction requires the infusion to be interrupted until symptoms resolve and then resumed at half the previous rate. A severe reaction requires the infusion to be stopped permanently, the subject to be treated according to local practice for anaphylaxis, and study treatment to be discontinued.

Appendix 12 PROHIBITED AND PERMITTED CONCOMITANT MEDICATION

Table 14 Concomitant medication

Category	Status	Comment
Other investigational agents	Prohibited	From 28 days before the first dose until the safety follow-up visit.
Other systemic therapy for the disease under study	Prohibited	During the treatment period; the start date of any subsequent therapy is recorded.
Live attenuated vaccines	Prohibited	From 28 days before the first dose until 90 days after the last dose.
Strong inhibitors or inducers of the principal clearance pathway	Avoid	Where unavoidable, monitor as described in Section 5.7 and record the interaction.
Medication known to prolong the corrected QT interval	Avoid	Where unavoidable, perform additional electrocardiograms.
Anticoagulants	Permitted	Monitor according to local practice; record any change in dose.
Corticosteroids	Permitted	Physiological replacement and short courses for the management of adverse events are permitted.
Growth factors	Permitted	According to local practice; not permitted for the sole purpose of meeting an eligibility criterion.
Antiemetics, analgesics and antimicrobials	Permitted	According to local practice.
Palliative radiotherapy	Permitted	To a site that is not used for disease assessment; study treatment is withheld during the course.

Appendix 13 GLOSSARY OF TERMS USED IN THIS PROTOCOL

Table 15 Glossary

Term	Definition
Adverse event	Any untoward medical occurrence in a subject administered a medicinal product, whether or not related to that product.
Adverse event of special interest	An event of scientific or medical concern specific to this product, requiring reporting to the sponsor within 24 hours whether or not it is serious.
Adverse reaction	An adverse event for which there is a reasonable possibility that the product caused it.
Analysis set	A defined group of subjects whose data contribute to a given analysis.
Assessment window	The period around a scheduled day within which an assessment may be performed without constituting a deviation.
Baseline	The last non-missing assessment obtained before the first dose of study treatment.
Blinding	A procedure in which one or more parties are unaware of the treatment assignment.
Case report form	The record on which the information required by the protocol is captured for each subject.
Censoring	The point at which a subject stops contributing information about whether an event has occurred.
Cohort	A group of subjects assigned to the same dose or treated under the same part of the study.
Comparator	The treatment against which the investigational product is compared.
Confirmed response	A response documented on two consecutive assessments separated by at least the interval specified in the protocol.
Cycle	The interval after which the pattern of study treatment administration repeats.
Data monitoring committee	An independent group that reviews accumulating data and advises the sponsor on continuation, modification or termination.
Database lock	The point after which no further change may be made to the study data without a documented and approved procedure.
Deviation	Any departure from the approved protocol.
Discontinuation	The permanent cessation of study treatment for a subject.
Dose-limiting toxicity	An event meeting the criteria in Section 7.3.1 during the defined observation period.
Dose intensity	The dose received in a period divided by the dose planned for that period.
End of study	The date defined in Section 3.4.
Enrolment	The point at which a subject is assigned to study treatment.
Estimand	A precise description of the treatment effect to be estimated, including the population, the variable, the handling of intercurrent events and the summary measure.
Full analysis set	All enrolled subjects, analysed according to the treatment assigned.
Good Clinical Practice	The international ethical and scientific quality standard for the design, conduct, recording and reporting of clinical studies.

Term	Definition
Important deviation	A deviation that could materially affect the completeness, accuracy or reliability of the data, or the rights, safety or wellbeing of a subject.
Informed consent	The process by which a subject voluntarily confirms willingness to take part, after being informed of all aspects relevant to that decision.
Intercurrent event	An event occurring after treatment starts that affects the interpretation or the existence of the measurement.
Interim analysis	Any analysis performed before the planned final analysis.
Investigator	The person responsible for the conduct of the study at a site.
Investigator brochure	The compilation of clinical and non-clinical data on the investigational product relevant to its study in subjects.
Life-threatening	An event which, in the view of the investigator, placed the subject at immediate risk of death at the time it occurred.
Lost to follow-up	A subject for whom the criteria in Section 6.3 have been met.
Medical monitor	The physician nominated by the sponsor to advise on the medical conduct of the study.
Missing data	A measurement that was intended to be collected but was not.
Multiplicity	The inflation of the type I error rate that arises when more than one comparison is made.
Non-serious	An adverse event that meets none of the seriousness criteria.
Objective response	A confirmed complete or partial response as defined by the response criteria in use.
Per-protocol set	The subset of the full analysis set without an important deviation affecting the primary endpoint.
Pharmacodynamics	The relationship between exposure to the product and its effect on the body.
Pharmacokinetics	The relationship between the dose administered and the concentration achieved over time.
Primary endpoint	The endpoint on which the principal conclusion of the study rests.
Progression	Disease worsening meeting the definition in the response criteria in use.
Randomisation	The allocation of subjects to treatment by chance.
Reference safety information	The section of the investigator brochure against which expectedness is assessed.
Relative dose intensity	Dose intensity expressed as a proportion of that planned.
Safety analysis set	All subjects who received at least one dose, analysed according to the treatment actually received.
Screen failure	A subject who consented but was not enrolled.
Serious adverse event	An adverse event meeting any criterion in Appendix 3.
Severity	The intensity of an event, graded as described in Appendix 7.
Source data	The original records, and certified copies of them, of clinical findings and observations.
Sponsor	The organisation that takes responsibility for the initiation, management and financing of the study.
Stratification	The division of the randomisation by prognostic factor so that treatment groups are balanced for it.

Term	Definition
Study treatment	Any product administered to a subject as required by this protocol.
Subject	An individual who takes part in a clinical study, whether as a recipient of the study treatment or as a control.
Suspected unexpected serious adverse reaction	A serious adverse reaction the nature or severity of which is not consistent with the reference safety information.
Treatment-emergent	An event that began, or worsened, on or after the first dose of study treatment.
Unexpected	Not consistent in nature or severity with the reference safety information.
Washout	A period without a given treatment before study treatment starts.
Withdrawal	The cessation of a subject's participation in the study.

Appendix 14 BLOOD VOLUME

Table 16 Indicative blood volume

Purpose	Volume per sample	Samples	Total
Haematology	4 mL	Per Table 5	Approximately 60 mL over the first 6 cycles
Clinical chemistry	6 mL	Per Table 5	Approximately 90 mL over the first 6 cycles
Coagulation	3 mL	Per Table 5	Approximately 24 mL over the first 6 cycles
Serology	6 mL	Screening only	6 mL
Pharmacokinetics	4 mL	Per Table 5	Approximately 40 mL, or 90 mL for a subject in the serial sampling subset
Biomarkers	10 mL	Per Table 5	Approximately 50 mL

The total volume drawn in any 8-week period does not exceed 450 mL. Where a subject is anaemic the investigator may omit an exploratory sample; omission of a safety sample is a deviation.

Appendix 15 COUNTRY-SPECIFIC REQUIREMENTS

Where the requirement of a territory is more restrictive than this protocol, the requirement of the territory applies and the difference is documented in a country-specific annex approved by the responsible authority and ethics committee before the first subject is enrolled in that territory. The following areas commonly differ and are addressed in the annexes: the age of majority for consent, the acceptable methods of contraception, the retention period for essential documents, the requirement to notify a subject's general practitioner, the arrangements for insurance and indemnity, and the language and format of the subject information sheet.

Consent by a legally designated representative. Some territories permit enrolment of a subject who cannot consent personally, through a legally designated representative, and others do not. Where it is permitted and the ethics committee approves it, the arrangements are set out in the country annex and the subject is consented personally as soon as they regain capacity.

Contraception. The methods that count as highly effective, and the period for which they are required, are set by the responsible authority in some territories. Where the local requirement is more restrictive than Appendix 4, the local requirement applies.

Notification of the general practitioner. Some territories require the subject's general practitioner to be informed of participation. Where that is required, the subject is told of it during consent and may decline participation on that basis.

Retention of essential documents. The retention period differs between territories. Where the local period exceeds that in Section 9.4, the local period applies and the site is told what it is before the study begins.

Insurance and indemnity. The form of cover required differs. Evidence of the cover in force in that territory is provided to the ethics committee before approval is sought.

Language and readability of subject documentation. The information sheet is provided in the language of the subject, at the reading level required locally, and the translation is certified. A back translation is provided to the sponsor where the ethics committee requires it.

Appendix 16 SUBJECT INFORMATION SHEET AND CONSENT FORM

This appendix is the master version. Each site prepares a local version in the language of the subject, with the local contact details and any wording required by the responsible ethics committee, and does not change the meaning of anything below.

16.1 Invitation

You are being invited to take part in a research study. Before you decide, it is important that you understand why the research is being done and what it would involve for you. Please take the time to read this information carefully. Talk to others about it if you wish. Ask us if anything is unclear or if you would like more information.

Taking part is entirely voluntary. If you decide not to take part, or to stop taking part later, your usual care will not be affected in any way and you do not have to give a reason.

16.2 Why is this study being done?

This study is being done to find out more about a medicine called Kelvaparib. Kelvaparib is being developed as a possible treatment for advanced solid tumours with a defect in one of the ways cells repair their DNA (homologous recombination), which is the condition you have.

We do not yet know whether Kelvaparib will help people with your condition. That is the question this study is designed to answer. It is possible that Kelvaparib will not help you, and it is possible that it will cause side effects.

About 58 people are expected to take part in this study at hospitals in several countries.

16.3 Why have I been invited?

You have been invited because your condition and your general health appear to match what the study requires. Before you can join, the study doctor will carry out some checks to confirm this. These checks are described below. Not everyone who has these checks will be able to join the study.

16.4 Do I have to take part?

No. It is up to you. If you decide to take part you will be asked to sign a consent form. You will be given a copy of this information sheet and of the signed form to keep.

You are free to stop at any time and without giving a reason. If you stop, we would like to keep the information we have already collected about you, and we would like to be able to find out how you are from time to time. You can say no to either of these and still stop taking the study medicine.

16.5 What will happen to me if I take part?

First there is a screening period of up to four weeks. During this time the study doctor will ask about your medical history and the treatments you have had, examine you, and arrange blood tests, a heart tracing and a scan. Women who could become pregnant will have a pregnancy test.

If the checks show that you can join, you will start the study medicine. Kelvaparib is given orally twice daily on a continuous schedule. You will come to the hospital for treatment and for checks according to a schedule the study doctor will explain and give you in writing.

At most visits we will examine you, measure your blood pressure, pulse and weight, ask how you have been feeling, ask about any other medicines you are taking, and take blood samples. Scans to see how your condition is responding will be done about every nine weeks at first.

You will stay on the study medicine for as long as it is helping you and you are not having side effects that cannot be managed. When you stop, we will ask you to come back for a final visit and then for a check-up about a month later. After that we would like to contact you every three months to ask how you are.

16.6 What will I have to do?

We will ask you to take the study medicine as instructed, to come to your visits, to tell us about any change in how you feel, and to tell us about any other medicine you start or stop, including medicines you buy yourself and herbal remedies.

There are some things you should not do during the study. You should not take part in another research study. You should not have a live vaccine. You should not donate blood. If you or your partner could become pregnant, you must use reliable contraception; the study doctor will explain which methods are acceptable and for how long.

16.7 What are the possible disadvantages and risks?

Like all medicines, Kelvaparib can cause side effects. The ones seen most often so far, and the ones we watch for most carefully, will be explained to you by the study doctor and are listed in a separate sheet that is updated as we learn more.

Because we do not yet know everything about this medicine, there may be side effects that have not been seen before. This is why we monitor you closely and why it is important to tell the study team about anything new, even if it seems minor.

Blood tests can cause discomfort and bruising. Scans involve exposure to a small amount of radiation; the amount is similar to what you would receive from natural sources over a period of a few years, and the study doctor can tell you more.

If you are pregnant, become pregnant, or are breastfeeding, you must not take the study medicine, because we do not know what effect it would have on a baby.

16.8 What are the possible benefits?

We cannot promise that the study will help you. Your condition may improve, stay the same or get worse while you are in the study, and we cannot tell in advance which it will be. The information we get from this study may help us to treat people with your condition in the future.

16.9 What if something goes wrong?

If you are harmed by taking part in this study, there are arrangements in place to compensate you. The study doctor will explain how these work where you live. If you are unhappy about any aspect of the study, please speak to the study team first. If you remain unhappy, you can complain through the hospital's usual complaints procedure.

16.10 Will my taking part be kept confidential?

Yes. Your name will not be recorded on any study document that leaves the hospital. You will be given a study number and that number will be used instead.

The study team, the company running the study and the authorities that regulate research may look at your medical records to check that the study is being carried out properly. Everyone who does so has a duty of confidentiality.

Information about you will be handled in accordance with the data protection law that applies where you live. You have rights over that information, and the study doctor can explain what they are and how to exercise them.

16.11 What will happen to my samples?

Blood samples taken for the study will be sent to a central laboratory. Samples are labelled with your study number, not your name. Some samples are used to measure how much of the medicine is in your blood; some are used for research into why the medicine works better in some people than others.

Samples that are left over may be stored for future research related to your condition or to this medicine. You can take part in the study and say no to this; there is a separate box on the consent form.

16.12 What will happen to the results?

The results will be published in a medical journal and on public registers of clinical research, whatever they show. You will not be identified in any report or publication. If you would like a summary of the results when they are available, please tell the study team and we will arrange it.

16.13 Who is organising and funding the study?

The study is organised and funded by Marnholt Oncology GmbH. The study doctor is not paid personally for including you; the hospital receives payment towards the cost of the extra work the study involves.

16.14 Who has reviewed the study?

All research in this hospital is looked at by an independent group of people, called a research ethics committee, to protect your interests. This study has been reviewed and approved by that committee and by the authority that regulates medicines where you live.

16.15 Consent form

Table 17 Statements to be initialled by the subject

	Statement	Initial
1	I confirm that I have read and understood the information sheet for this study and have had the opportunity to ask questions.	
2	I understand that my participation is voluntary and that I am free to withdraw at any time, without giving a reason and without my medical care or legal rights being affected.	
3	I understand that the study doctor may stop my treatment at any time if it is in my interests.	
4	I agree to my medical records being looked at by the study team, the sponsor and the regulatory authorities, where it is relevant to my taking part.	
5	I agree to my general practitioner being informed that I am taking part.	
6	I agree to blood samples being taken and used as described in the information sheet.	
7	I agree to my leftover samples being stored for future research related to my condition or to this medicine.	
8	I agree to be contacted for follow-up after I stop taking the study medicine.	
9	I agree to take part in the above study.	

Item 7 is optional. A subject may decline it and still take part.

Table 18 Signatures

Role	Name (print)	Signature	Date
Subject			
Person taking consent			
Investigator			

Signature fields are left blank.